Valuation Renewal, Finite Residue Connectivity, and the Orbitwise Gap in Collatz Dynamics

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Abstract

We develop a structural analysis of the Collatz map through the lens of renewal theory, 2-adic symbolic dynamics, and finite residue control. The central result is that divergence of a Collatz orbit, if it exists, must simultaneously satisfy three conditions: the empirical valuation average falls below $\log_2 3$, the valuation complexity is unbounded ($\sup_r k_r = \infty$), and the orbit sustains orbitwise arithmetic decorrelation that escapes the finite-information renewal structure.

Concretely, we establish: (1) **Theorem AV**: under the Haar/renewal population cycle model, the average odd-step valuation equals $2 > \log_2 3$, giving a negative population Lyapunov drift; (2) **Theorem NRC**: the mod-2^K Type A residue transition graph G_K is strongly connected for all $K \geq 3$, via explicit valuation pattern constructions (Lemmas VP and TC); (3) **Theorem BVC**: bounded valuation complexity implies eventual finite symbolic periodicity, so a divergent orbit must have unbounded k_r.

The paper does not prove the Collatz conjecture. Rather, it identifies with precision the structural gap between population renewal averages and orbitwise ergodic behavior, and locates the residual obstruction in an orbitwise arithmetic decorrelation problem naturally connected to Chowla-type conjectures.

Contents

1 Introduction **2**

2 Type A / Type B Cycle Structure **2**
2.1 Definitions and basic coding 2
2.2 One-cycle decomposition 2

3 Population Drift **3**

4 Finite Residue Connectivity **3**
4.1 The transition graph 3
4.2 Valuation pattern realizability 4
4.3 Target control 4
4.4 Strong connectivity 4

5 Bounded Valuation Complexity **5**

6 The Orbitwise Gap and Open Problems **5**
6.1 The hierarchy of conditions 5
6.2 What NRC does and does not give 6
6.3 The ergodic lift (conditional) 6
6.4 The arithmetic obstruction (OP4-Chowla) 6
6.5 Summary of open problems 6

7 Conclusion **7**

1 Introduction

The Collatz map $T : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ is defined by $T(n) = n/2$ for even n and $T(n) = (3n+1)/2^{v_2(3n+1)}$ for odd n, where v_2 denotes the 2-adic valuation. The Collatz conjecture asserts that every orbit eventually reaches 1.

A standard heuristic observes that odd iterates satisfy $E[\log n_{t+1}/n_t] = \log(3/4) < 0$ on average, suggesting orbits shrink. Yet this population-level observation does not rule out individual divergent orbits. The gap between these two levels is the subject of this paper.

Main message. We establish that

$$\text{population shrinkage} \neq \text{orbitwise convergence},$$

and make this gap precise through three complementary results: a population renewal theorem (Theorem AV), a finite residue connectivity theorem (Theorem NRC), and a bounded-complexity periodicity theorem (Theorem BVC). Together, these results show that a hypothetical divergent orbit must evade a finite-information regenerative mechanism by sustaining unbounded valuation complexity and orbitwise arithmetic coherence.

Relation to companion papers. This paper builds on the exact operator structure established in Kamanoi (2026a), which proved that the renewal shift carries total mutual information $\sum_{N\geq 1} I_\pi(k_0; k_N) = 2\log 2$ (finite), and on Kamanoi (2026b), which showed that divergence implies atypicality ($\mu_n \neq \pi$) and a thermodynamic obstruction. The present paper develops the finite-residue and valuation-complexity sides of the problem.

2 Type A / Type B Cycle Structure

2.1 Definitions and basic coding

Let $\mathcal{A} = \{n \in \mathbb{Z}_{>0} : n \equiv 3 \pmod 4\}$ be the set of *Type A* integers, and let $T_{\mathcal{A}}$ denote the first-return map to $\mathcal{A}$. For $n \in \mathcal{A}$, define the *coding*

$$k(n) = v_2(n+1) - 1 \geq 1,$$

which takes values in $\{1, 2, 3, \ldots\}$ and governs the renewal structure.

Integers with $n \equiv 1 \pmod 4$ are *Type B*; they satisfy $v_2(3n+1) \geq 2$.

2.2 One-cycle decomposition

Each Type A return cycle decomposes into two phases:

Type A block. Starting from $n_0 \in \mathcal{A}$ with $k(n_0) = j$, Lemma A of Kamanoi (2026a) gives a deterministic descent:

$$n_0 \xrightarrow{v=1} n_1 \xrightarrow{v=1} \cdots \xrightarrow{v=1} n_{j-1},$$

where $k(n_s) = j - s$ and each step has $v_2(3n_s + 1) = 1$. After j steps, the orbit exits to a Type B point.

Type B excursion. From the Type B exit point, the orbit wanders among Type B points until returning to $\mathcal{A}$. The excursion length ℓ and the valuations $v_s \geq 2$ at each step are governed by the following.

Lemma 2.1 (B-excursion)**.** *For $n \equiv 1 \pmod 4$ under the Haar population measure:*

(i) $P(v_2(3n+1) = s \mid n \equiv 1 \pmod 4) = 2^{-(s-1)}$ *for* $s \geq 2$.

(ii) The next odd iterate $n' = (3n+1)/2^{v_2(3n+1)}$ *satisfies* $P(n' \equiv 3 \pmod 4) = P(n' \equiv 1 \pmod 4) = 1/2$.

(iii) $v_2(3n+1)$ *and* $n' \bmod 4$ *are independent.*

Hence $\ell \sim \mathrm{Geom}(1/2)$ *independently of the Type A block, and* $E[v \mid \text{Type } B] = 3$.

3 Population Drift

Let V denote the total odd-step valuation sum over one cycle, and L the total number of odd steps.

Theorem 3.1 (Average Valuation)**.** *Under the Haar/renewal population cycle model,*

$$\frac{\mathbb{E}[V]}{\mathbb{E}[L]} = 2 > \log_2 3 \approx 1.585.$$

Proof. Write $j = k(n_0) \sim \mathrm{Geom}(1/2)$, so $\mathbb{E}[j] = 2$.

Type A block: j steps, each with $v = 1$. Contribution: $\mathbb{E}[V_A] = 2$, $\mathbb{E}[L_A] = 2$.

Type B excursion: by Lemma 2.1, $\ell \sim \mathrm{Geom}(1/2)$ with $\mathbb{E}[\ell] = 2$, and $E[v \mid B] = \sum_{s=2}^{\infty} s \cdot 2^{-(s-1)} = 3$. Since j, ℓ, and the individual v_s are independent, Wald's identity gives

$$\mathbb{E}[V_B] = \mathbb{E}[\ell] \cdot E[v \mid B] = 6, \qquad \mathbb{E}[L_B] = \mathbb{E}[\ell] = 2.$$

Therefore $\mathbb{E}[V] = 8$, $\mathbb{E}[L] = 4$, and the ratio is 2. □

Corollary 3.2 (Population Lyapunov Drift)**.**

$$\lambda_{\mathrm{pop}} = \frac{\mathbb{E}[V]}{\mathbb{E}[L]} \cdot (-\log 2) + \log 3 = \log \frac{3}{4} < 0.$$

Population orbits shrink on average.

Remark 3.3 (Remaining gap)**.** *Theorem 3.1 is a* population *statement. The passage to individual orbits requires a cycle-level law of large numbers:*

$$\liminf_{J \to \infty} \frac{\sum_{r<J} V_r}{\sum_{r<J} L_r} > \log_2 3,$$

which is not automatic from population averages alone. This is the orbitwise gap addressed in Section 6.

4 Finite Residue Connectivity

4.1 The transition graph

Fix $K \geq 3$. Define the *Type A residue classes* modulo 2^K:

$$\mathcal{C}_K = \{a : 1 \leq a < 2^K,\ a \equiv 3 \pmod 4\}, \quad |\mathcal{C}_K| = 2^{K-2}.$$

The *cycle-return graph* G_K has vertex set $\mathcal{C}_K$ and a directed edge $a \to b$ whenever $P_K(a, b) > 0$, where $P_K(a, b)$ is the probability that a Type A orbit starting in class a first returns to class b (modulo 2^K).

4.2 Valuation pattern realizability

Lemma 4.1 (VP: Valuation Pattern Realizability)**.** *For any finite sequence $(v_1, \ldots, v_\ell)$ with $v_s \geq 1$ and any odd integer n_ℓ, define $V_\ell = \sum_{s=1}^{\ell} v_s$ and the constant c_ℓ by $c_0 = 0$, $c_s = 3c_{s-1} + 2^{V_{s-1}}$. Then*

$$n_0 \equiv 3^{-\ell}(2^{V_\ell} n_\ell - c_\ell) \pmod{2^{V_\ell+\ell}}$$

is the unique residue class modulo $2^{V_\ell+\ell}$ realizing the valuation pattern $(v_1, \ldots, v_\ell)$ starting from n_0 and reaching n_ℓ.

Proof. One checks by induction that $n_s = (3^s n_0 + c_s)/2^{V_s}$ and that $v_2(3^s n_0 + c_s) = V_s$ exactly (so n_s is odd at each step), using the recurrence for c_s and the fact that 3 is a 2-adic unit. □

4.3 Target control

Lemma 4.2 (TC: Target Control)**.** *For any $K \geq 3$ and any $b \equiv 3 \pmod 4$, setting $m = \lceil (K-1)/2 \rceil$ and taking the pattern $(v_1, \ldots, v_{m+1}) = (1, \underbrace{2, \ldots, 2}_{m})$ yields an initial class*

$$n_0 \equiv 3^{-(m+1)}(2^{1+2m} b - c_{m+1}) \pmod{2^{3m+2}}$$

satisfying $n_0 \equiv 3 \pmod 8$ (the $k = 1$ class) and $n_{m+1} \equiv b \pmod{2^K}$.

Proof. We verify $n_0 \equiv 3 \pmod 8$ in both parities of m.

$c_{m+1} \bmod 8$. For the pattern $(1, 2, 2, \ldots)$, $V_{s-1} = 2s - 3 \geq 3$ for $s \geq 3$, so $2^{V_{s-1}} \equiv 0 \pmod 8$ and $c_s \equiv 3c_{s-1} \pmod 8$ for $s \geq 3$. With $c_1 = 1$ and $c_2 = 5$, this gives the period-2 pattern: $c_s \equiv 5 \pmod 8$ for s even, $c_s \equiv 7 \pmod 8$ for s odd ($s \geq 2$).

$3^{-(m+1)} \bmod 8$. Since $3^2 \equiv 1 \pmod 8$, we have $3^{-k} \equiv 1 \pmod 8$ for k even and $3^{-k} \equiv 3 \pmod 8$ for k odd.

Case m odd ($m+1$ even): $3^{-(m+1)} \equiv 1$, $c_{m+1} \equiv 5$, so $n_0 \equiv -5 \equiv 3 \pmod 8$. ✓

Case m even ($m+1$ odd): $3^{-(m+1)} \equiv 3$, $c_{m+1} \equiv 7$, so $n_0 \equiv -21 \equiv 3 \pmod 8$. ✓

Since $1 + 2m \geq K$, the term $2^{1+2m} b$ carries the full $b \bmod 2^K$ information, giving the target control. □

4.4 Strong connectivity

Theorem 4.3 (NRC: Non-trivial Residue Connectivity)**.** *For all $K \geq 3$, the graph G_K is strongly connected.*

Proof. **Claim B** (any class $\to$ $k = 1$ class). For any $a \in \mathcal{C}_K$ with $k(a) = j$, Lemma A of Kamanoi (2026a) gives a deterministic j-step descent to a class with $k = 1$ (i.e., $\equiv 3 \pmod 8$).

Claim A ($k = 1$ class $\to$ any class). Lemma 4.2 provides, for every target $b \in \mathcal{C}_K$, an explicit valuation pattern realizing $n_{m+1} \equiv b \pmod{2^K}$ from a $k = 1$ starting class.

Claims A and B together imply strong connectivity. □

Remark 4.4 (NRC vs. Doeblin)**.** *Strong connectivity of G_K gives* finite irreducibility*: a directed path exists from any class to any other. It does not automatically yield a Doeblin minorization $P_K^{m_K}(a, \cdot) \geq \varepsilon_K \pi_K(\cdot)$, which requires a uniform lower bound on path probabilities. This stronger condition— finite irreducibility with uniform path probability— is the bridge to an ergodic lift and remains an open step; see Section 6.*

5 Bounded Valuation Complexity

Theorem 5.1 (BVC: Bounded Valuation Complexity)**.** *If* $\sup_r k_r \le M < \infty$*, then the induced symbolic residue orbit* $(n_r \bmod 2^{M+2})_{r\ge 0}$ *is eventually periodic.*

Proof. Since $k_r \le M$, we have $v_2(n_r+1) \le M+1$, so $n_r \bmod 2^{M+2}$ lies in the finite set

$$\mathcal{C}_M^* = \{a : a \equiv 3 \pmod 4,\ v_2(a+1) \le M+1\} \subset \{1, \dots, 2^{M+2}-1\}.$$

The Type A return map sends $\mathcal{C}_M^*$ into itself: if $k(n) \le M$, then $k(T_{\mathcal{A}}(n)) \le M$ follows from the renewal descent (Lemma A). A deterministic map on a finite set is eventually periodic. □

Corollary 5.2 (Growth obstruction)**.** *If the orbit is eventually periodic with period* P*, then* $n_{r+P}/n_r \to 3^{L_P}/2^{V_P}$ *as* $n_r \to \infty$*, where* L_P, V_P *are the total odd-step count and valuation sum over one period, both determined by the finite residue cycle. Divergence requires* $3^{L_P} > 2^{V_P}$*, i.e.,*

$$L_P/V_P > \log_3 2 \approx 0.631.$$

Corollary 5.3 (Bounded k and nontrivial cycles)**.** *If no nontrivial Collatz cycle exists (verified computationally for* $n \le 2^{68}$*), then* $\sup_r k_r \le M$ *implies* $n_r \to 1$*.*

Corollary 5.4 (Divergence dichotomy)**.** *A divergent Collatz orbit must satisfy either:*

(a) $\sup_r k_r = \infty$ *(unbounded valuation complexity), or*

(b) $\sup_r k_r \le M$ *and the symbolic residue cycle grows (i.e.,* $3^{L_P} > 2^{V_P}$*, which requires a nontrivial cycle).*

6 The Orbitwise Gap and Open Problems

6.1 The hierarchy of conditions

The preceding results establish the following implication chain:

$$\text{OP4-full} \Rightarrow \text{cycle-level LLN} \Rightarrow \text{OP4-min} \Rightarrow \text{no divergence},$$

where:

$$\begin{aligned}
\text{OP4-min}: &\quad \liminf_{M\to\infty} \frac{1}{M} \sum_{t<M} v_2(3n_t+1) > \log_2 3, \\
\text{OP4-mean}: &\quad \frac{1}{M} \sum_{t<M} k_t \to 2, \\
\text{OP4-full}: &\quad \frac{1}{M} \sum_{t<M} \mathbf{1}_{k_t=j} \to 2^{-j} \quad \forall\, j \ge 1.
\end{aligned}$$

Theorem 3.1 establishes OP4-min at the *population* level. Theorem 4.3 establishes that the finite residue structure is irreducible. Neither result gives the orbitwise statement directly.

6.2 What NRC does and does not give

NRC gives *finite irreducibility*: for every K and every pair of residue classes $a, b \in \mathcal{C}_K$, there exists a finite path from a to b in G_K. This is strictly weaker than the Doeblin minorization

$$P_K^{m_K}(a, \cdot) \geq \varepsilon_K \pi_K(\cdot),$$

which would give uniform mixing and, via a coupling argument, ergodicity of the cycle process (V_r, L_r).

6.3 The ergodic lift (conditional)

Theorem 6.1 (EL: Ergodic Lift, conditional)**.** *Assume finite-level Doeblin: for every K, there exist $m_K \geq 1$ and $\varepsilon_K > 0$ such that $P_K^{m_K}(a, \cdot) \geq \varepsilon_K \pi_K(\cdot)$ for all a. Then the cycle process $(V_r, L_r)_{r \geq 1}$ is ergodic, and*

$$\frac{1}{J} \sum_{r<J} V_r \to 8, \qquad \frac{1}{J} \sum_{r<J} L_r \to 4 \quad \text{a.s.}$$

Consequently,

$$\frac{\sum_{r<J} V_r}{\sum_{r<J} L_r} \to 2 > \log_2 3,$$

which implies OP4-min *for every Collatz orbit and excludes divergence.*

6.4 The arithmetic obstruction (OP4-Chowla)

The Doeblin condition requires that the return probabilities $P_K(a, b)$ are bounded away from zero uniformly. This in turn requires that the Type B excursion return map mixes the residue classes.

Concretely, OP4-full is connected to the following decorrelation condition for the arithmetic weight $\Lambda_C(n) = \mathbf{1}_{(3n+1)/2 \text{ prime}}$:

$$\frac{1}{M} \sum_{t<M} \Lambda_C(n_t) \Lambda_C(n_{t+h}) \to \left(\frac{1}{M} \sum_{t<M} \Lambda_C(n_t) \right)^2 \quad \forall\, h \neq 0.$$

This is the Chowla conjecture for Λ_C, and controlling it requires parity information on $\Omega((3n+1)/2) \bmod 2$—precisely the *sieve parity barrier*. Standard sieve methods are insensitive to this parity, and no current technique breaks this barrier.

Remark 6.2 (Connection to Chowla)**.** *OP4-full is* naturally connected *to Chowla-type decorrelation for Λ_C and appears to meet the sieve parity barrier. We do not claim equivalence; the connection is structural and the difficulty is at least comparable.*

6.5 Summary of open problems

Level	Statement	Status
OP4-min	$\liminf \frac{1}{M} \sum v_t > \log_2 3$	**Open (orbitwise)**
OP4-mean	$\frac{1}{M} \sum k_t \to 2$	**Open**
OP4-full	$\frac{1}{M} \sum \mathbf{1}_{k_t = j} \to 2^{-j}$	**Open**
OP4-Chowla	Orbitwise decorrelation of Λ_C	**Open (parity barrier)**
Doeblin	$P_K^{m_K}(a, \cdot) \geq \varepsilon_K \pi_K(\cdot)$	**Open**
Trivial cycles	Bounded orbit $\Rightarrow n_r \to 1$	Verified $\leq 2^{68}$

7 Conclusion

We have shown that the Collatz problem, when viewed through the renewal and residue-connectivity lens, reduces to a single orbitwise question:

Can a positive-integer Collatz orbit sustain an arithmetic deviation from the 2-adic renewal law?

The results of this paper narrow the structural constraints on any hypothetical divergent orbit as follows. Such an orbit must:

1. have empirical valuation average at most $\log_2 3$ (D1-bwd, Kamanoi 2026b),
2. differ from the stationary renewal measure π (NGM, Kamanoi 2026b),
3. have unbounded valuation complexity $\sup_r k_r = \infty$ (Theorem 5.1),
4. avoid all finite residue classes being visited uniformly, despite the graph G_K being strongly connected (Theorem 4.3),
5. sustain long-range arithmetic correlations that escape the finite-memory renewal background (OP4-Chowla).

The framework therefore suggests that the unresolved core of the Collatz problem is not the average drift—which is negative at the population level—but the possibility of orbitwise arithmetic coherence persisting indefinitely against an exact renewal background.

References

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